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Design and Implementation of Intelligent Software Version Management Platform for Power Communication Network Based on Large Language Models

Jianing Niu[1], Longchuan Yan[1], Ziyan Zhao[1], Yongyuan He[1], Ning Zhang[1], Guilin Song[1], Xiaoyu Yuan[2*]
1. State Grid Information&Telecommunication Branch, Beijing, China
2. Beijing Kedian Tuoyu Intelligent Technology Co., Ltd, Beijing, China
jianing-niu@sgcc.com.cn, 49864286@qq.com, 150382031@qq.com, 5783016@qq.com, 623860762@qq.com, 873905548@qq.com, michaelyuanxiaoyu@qq.com
Corresponding Author: Xiaoyu Yuan Email: michaelyuanxiaoyu@qq.com

***Abstract*—With the rapid development of power communication network, software version update have become a heavy task in network operation and maintenance. It's necessary to manage software versions on an automated and intelligent platform to facilitate IT engineers updating software. In this paper, we propose a design and implementation of intelligent software version management platform based on Large Language Models (LLMs). Evaluation shows that LLMs can play an important role in software updating process and this intelligent platform can provide assistance to operations and maintenance of power communication network.**

Keywords—power communication network; Large Language Models; software version management; software update

I. INTRODUCTION

The power communication network is a kind of network used for information transmission and exchange within the power system, covering all aspects of power production, distribution, and consumption. It is an important means to ensure the safe, stable, and economic operation of the power grid. Therefore, the safe and stable operation of the power communication network is crucial for the normal operation of the power grid, and the normal operation of communication equipment is the foundation of the operation of the power communication network.

In recent years, with the rapid expansion of power communication network and the increase in device types, network operation and maintenance tasks have become increasingly heavy. Among them, software version management of network device has gradually become a huge challenge[1]. The acceleration of information technology has made software version updates of communication network more frequently. Traditional manual management methods are not only inefficient, but also prone to risks, making it difficult to meet the needs of modern power communication network management for massive devices and software.

Updating device software version is a common task encountered by operation and maintenance engineers. Traditionally, engineers evaluate the various risks that a new version of software may bring to the device and the network based on the information released by the equipment manufacturer, and then decide whether to update the new version. This approach relies entirely on the abilities and experience of engineers, which can easily lead to new problems and may bring disasters to communication network. There is an urgent need for intelligent methods to assist software version updates and reduce human factors. At the same time, Large Language Models (LLMs) have developed rapidly in recent years and have had a huge impact and application value in various fields[2][3][4][5]. Undoubtedly, in the field of IT operations, especially software version updates, there is also a certain potential for the application of LLMs, which is worth researching and trying.

Facing the growing challenges of software version management, it has a great value to develop an automated and intelligent network device version management and risk assessment platform that utilizes the reasoning ability of LLMs to empower the operation and maintenance of power communication network, further ensuring the safe, stable, and efficient operation of power communication networks.

II. BACKGROUND AND RELATED WORKS

In recent years, the integration of technologies such as the Internet of Things, big data, and cloud computing has provided new possibilities for the intelligent operation and maintenance management of power communication network. However, there is relatively little research on software version management for power communication network, especially on how to achieve efficient and secure version updates while

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ensuring stable network operation, which still faces many challenges.

Updating the version of power communication network equipment is an important process involving the safe and stable operation of power communication systems. This process requires strict adherence to prescribed procedures to ensure uninterrupted communication, data loss, and system security and stability during the update process.

In general, the basic process of updating the version of power communication network equipment mainly includes the following steps[1][7][8]:

(1) Requirements Analysis: Analyzing updates requirements includes determining the objectives of the software update, defining the scope to update, identifying the new version, understanding any dependencies and potential conflicts, assessing impacts on devices and networks, and so on.

(2) Update Planning: This step includes identifying the environments of hardware and software, making sure the new software version to upgrade ready, determining the every operations to execute, selecting an optimal time to operate updates, verifying the functionalities and performance of devices and networks after updating, and so on.

(3) Testing Before Updating: Before updating, the new version software should be tested in a simulation environment or testing environment comprehensively.

(4) Update Preparation: This step is to prepare all necessary tools in place and make sure all devices in networks accessible and ready for the update.

(5) Update Distribution: This stage is executing operations of update plan step-by-step to install the new version or patch of software on all relevant network devices.

(6) Verification: After updating, it's better to instantly verify all software functions working normally, to monitor performance of the networks to show the benefits of the updates, and to re-evaluate the security of the networks comprehensively.

In the entire software version update process, security and stability are the primary considerations. Each step needs to be strictly evaluated and executed according to the operating procedures to ensure the reliable operation of the power communication network. At the same time, it is necessary to prepare emergency plans and backup old versions and data during the update process to cope with possible emergencies.

In past several years, Transformer[6] based large language models have developed rapidly and demonstrated extremely superior processing capabilities in natural language processing tasks. Research has shown that very large scale models not only improve performance, but also exhibit unique abilities such as context learning and emergence. With the expansion of training data, continuous innovation of training methods, and the increasingly complex structure of LLMs, these models have been widely applied in multiple industries such as education, finance, healthcare, and industry[2][3][5][9]. Meanwhile, many scholars are also studying the application of large models in the field of intelligent operation and maintenance[10]. The basic large language model is trained using general language corpus, which is difficult to apply in the highly specialized IT operations field. Therefore, using knowledge from the IT operations field to fine-tune big models is a new research field and application direction.

DevOps-Model[12] is a large language model jointly released by Ant Group and Peking University for the chinese DevOps field. DevOps-Model collects professional data related to the DevOps field, and then conducts language model training and alignment training on the model to produce large models that can help engineers improve efficiency throughout the entire development and operation life-cycle.

OWL[13] is a large model fine-tuned based on professional knowledge in the field of operations and maintenance. OWL uses self-instruct strategy to collect datasets. The datasets includes 9 common domains, such as information security, system architecture, network operation and maintenance, operating system operation and maintenance, infrastructure operation and maintenance. It contains various IT related tasks, improving LLMs' generalization ability for IT operations. OWL has also built the Owl Bench evaluation benchmark datasets and introduced a mix of adapter strategies to further improve instruction tuning performance. The extensive experimental results conducted on Owl Bench have demonstrated the effectiveness of Owl in IT operations.

III. System Requirement Analysis

In the operation and maintenance of power communication network, operation and maintenance engineers usually need to record the software and hardware version information on each device, and modify the information according to each software and hardware update to ensure that the recorded information is consistent with the actual situation. After device manufacturers release a new version, they usually notify users to follow the latest version released on their official website through phone calls, emails, and other means. Users who receive the notification can immediately obtain information about the new version. However, there are many users of the device, and device manufacturers often cannot notify every user in a timely manner. These users who have not received notifications need to constantly pay attention to the device manufacturer to avoid missing important versions released by the device manufacturer and causing risks or losses. After learning about the release of a new version of the manufacturer's equipment,

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network operation and maintenance engineers need to visit the official release address to view the information of the new version, evaluate the pros and cons of the update based on the current network status, and then decide whether to download the new version and when to update the software.

In actual production, there are numerous types and quantities of power communication network equipment, and limited engineers in the operation and maintenance department find it difficult to know the status of each device. Therefore, a device information management system is needed to uniformly manage various information of devices in the network. Operations engineers manually input equipment information or automatically access various types of information through programs to obtain the required information, and update it in a timely manner according to changes in the equipment environment. Similarly, there are numerous device manufacturers, and operation and maintenance engineers need to spend a lot of energy paying attention to the release of new versions of devices. The channels for device manufacturers to release new versions are often fixed, so it is necessary and feasible to use automated methods to obtain the release information of new versions of devices.

After obtaining the latest version update information of the equipment, the operation and maintenance engineer needs to evaluate the value and risks brought by the version upgrade. When the value brought by the new version is minimal, it may be considered not to update temporarily. When a new version has a certain value that can be updated or must be updated and will bring huge risks, it is necessary to carefully evaluate the various risks faced during the update process.

If the risk assessment determines the need for a new version upgrade, a detailed upgrade strategy needs to be developed to ensure the smooth completion of the upgrade process. The upgrade strategy generally includes: preparation before upgrade, upgrade operation, post upgrade inspection, version rollback, fault handling, etc.

After assessing risks and developing upgrade strategies, the system can issue upgrade notifications to remind operations engineers to perform version upgrades.

In order to meet the needs of tracking software and hardware version information of power communication network equipment, discovering new versions, assessing update risks, and releasing new versions more efficiently, it is necessary to build an intelligent management platform that includes the following main functions:

(1) Device Information Management: Obtain the current version, environment, and other information of the device.

(2) Version Intelligent Management: Automatically obtain the latest version information provided by equipment suppliers, automatically download it to the platform, and make it available for platform users to download.

(3) Risk Evaluation: Evaluate the risks of version upgrade, develop version upgrade strategy recommendations, operational steps, precautions, etc.

(4) Updates Release: Release version upgrade reminder.

The values that the intelligent management platform for software versions of power communication networks can bring is as below:

(1) Process Standardization: The platform can standardize the management of software version update processes to ensure the compliance of update operations.

(2) Intelligent Management: The platform can achieve intelligent software version updates, improve update efficiency, and reduce manual intervention.

(3) Risk Control: The platform can assess and control the risks during the software version update process to ensure the stable operation of the power communication network.

IV. Systems Design

A. System Architecture

This system adopts a microservice architecture, which is divided into a data storage layer, a data collection layer, an analysis and decision-making layer, and a user interaction layer. The data storage layer is responsible for storing basic data such as device information, software new version information, and software new version files. The data collection layer automatically collects device information through protocols such as SNMP and NETCONF, and obtains software version update information through web crawlers. The analysis and decision-making layer is responsible for version updates risk assessment, development of update and upgrade strategies, and issuing version update reminders. The user interaction layer provides a visual operation interface.

B. Key Technologies

The key technologies involved in this system mainly include data collection technology and LLMs technology.

Data collection includes two types: device information collection and software new version information collection. Device information collection is achieved through network protocols such as SNMP and NETCONF[11]. The collection of software new version information is automatically carried out through web crawlers to the website where the device manufacturer releases the new version, and the relevant documentation, downloadable new version files, and

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other materials related to the new version are downloaded to the platform storage management.

The LLMs technology is using LLMs to assess the risk of whether to update the device with a new software version based on the current information of the device, software new version information, etc. For the new version updated through risk assessment, the system will develop corresponding upgrade strategies using a LLM, including upgrade operation steps, data backup, post upgrade checks, and other matters.

V. Implementation and Evaluation

A. System Implementation

The intelligent management platform for power communication network software versions is developed using Python, with Flask framework as the development foundation, and MySQL database for data storage. The platform interface is developed using the Vue.js framework for achieving a good user experience. The tasks of version upgrade risk assessment and upgrade strategy formulation are implemented using open-source LLMs and development frameworks.

Considering that new software versions are constantly generated over time, large models often cannot immediately learn relevant knowledge and have certain limitations. Therefore, it is necessary for device manufacturers to provide various information related to the new software version to the large model as a reference for inference. Here, we use the large model in two ways: one is to directly call the large language model to answer, and the other is to use retrieval enhanced generation (RAG)[14] to answer.

Due to the fact that the large language model is trained on large-scale general data, there is limited knowledge in the vertical domain, which can easily lead to hallucinations and incorrect results. We chose to call the DevOps-Model, a large language model that has been fine-tuned in the IT operations field, to reduce the occurrence of hallucinations.

The combination of retrieval enhanced generation (RAG) and prompt engineering[15] is safer. RAG is a hybrid mechanism based on information retrieval and content generation, which can ensure that the content generated by LLM is more accurate, professional, and rich by extracting question related information from knowledge base files. RAG typically first converts documents into vector embeddings, and then stores these vectors as a knowledge base. Based on this vector database, we use LLM to conduct dialogues and summaries on these processed documents. On the other hand, prompt engineering is also a crucial step in LLM development, as it strengthens the input of large models, helps LLM better understand human needs, and produces more accurate, relevant, and informative outputs. On the RAG framework, we compared two open-source projects, Langchain-Chatchat and QAnything. Langchain-Chatchat has implemented basic RAG functionalities, and the underlying model uses GLM-4-9B. QAnything has optimized many aspects and adopted QWenLM as the underlying model.

B. Evaluation

The core function of this system is to use a large language model for software version update risk assessment, develop upgrade strategies, etc. Here, we use the fine-tuned DevOps-Model, Langchain-Chathat[16] based on GLM-4-9B[18], and QAnything[17] based on QWenLM[19] to ask questions related to version updates to verify the effectiveness. The results are shown as Fig.1.

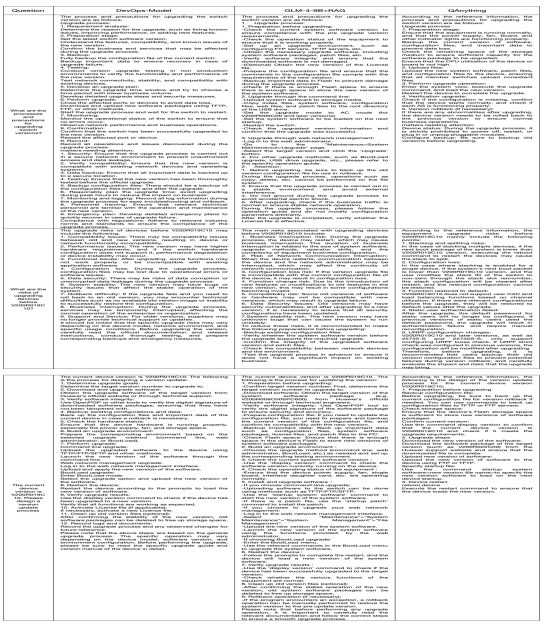


Fig. 1. Questions and answers for evaluation

DevOps-Model has made domain adjustments using IT operation and maintenance data. Its answers to communication network equipment software version updates are relatively generic and lack professionalism. Langchain-Chathat and QAnything answered questions based on the uploaded version documentation, and their accuracy is relatively high, with certain reference value.

In summary, due to the delayed knowledge of large models and the hallucination problem of LLMs, using fine-tuned models for software version update risk assessment and development of upgrade strategies has a high level of risk, while the RAG-based approach can more accurately meet the needs of operation and maintenance engineers for version upgrade related issues and has high feasibility.

VI. Conclusion

This paper proposes an intelligent software version management design scheme using LLMs to address the problems in software version management of power communication network. It can manage device software versions, assess risks, and recommend upgrades. This

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solution can effectively solve the inefficiency and insecurity problems of traditional management methods, improve the efficiency of network operation and maintenance, and enhance the intelligence level of software version management. This platform improves the efficiency of software version updates, reduces the risks associated with updates, and ensures the stable operation of the power communication network through intelligent and automated management of the software version update process. In future work, we will further improve the platform functionality, enhance the application of LLMs in version management and updates, and improve the compatibility and scalability of the platform to meet the constantly evolving needs of power communication network operation and maintenance.

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REFERENCES

[1] H. Yuan, R. Dou, S. Liu, et al. "Design Scheme for Version Control System of Intelligent Electronic Equipment in Substation[J]". Power System Automation, 2018, 042 (024): 163-168

[2] Naveed, Humza & Khan, Asad & Qiu, Shi & Saqib, Muhammad & Anwar, Saeed & Usman, Muhammad & Barnes, Nick & Mian, Ajmal. (2023). "A Comprehensive Overview of Large Language Models," arXiv:2307.06435v9, 2024.

[3] C. Huang, S. Li, R. Liu, H. Wang, Y. Chen, "Large Foundation Models for Power Systems," arXiv:2312.07044v1, 2023.

[4] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "Llama: Open and efficient foundation language models," CoRR, abs/2302.13971, 2023.

[5] S. Wu, O. Irsoy, S. Lu, V. Dabravolski, M. Dredze, S. Gehrmann, P. Kambadur, David S. Rosenberg, and G. Mann. "Bloomberggpt: A large language model for finance," CoRR, abs/2303.17564, 2023.

[6] Vaswani, Ashish, Noam M. Shazeer, N. Parmar, J. Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser and Illia Polosukhin. "Attention is All you Need," Neural Information Processing Systems , 2017.

[7] B. Cao, Z.W. Luan, D. Luo, et al. "Design and Implementation of a Scale Up System for Communication Modules in the Power Internet of Things," Rural Electrification, 2023 (9): 33-37. DOI: 10.13882/j.cnki. ncdqh. 2023.09.008

[8] Y. Jia, B. Shan, H. Dong. "Organization and Implementation of Software Version Upgrade for Large scale Programmable Switches," Computer and Information Technology, 2006 (11): 3. DOI: CNKI: SUN: JISJ.0.2006-11-036

[9] Y. Wang, Q. Li, Z. Dai , Y. Xu. "Current status and trends in large language modeling research," Chinese Journal of Engineering, 2024, 46(8):1411-1425. doi: 10.13374/j.issn2095-9389.2023.10.09.003

[10] D. Pei, S. Zhang, Y. Sun, et al. "Artificial intelligence for IT operations in era of large language model," ZTE technology journal, 2024, 30(2): 56-62. DOI: 10.12142/ZTETJ.202402009

[11] Y. Zeng, B. Lin, Z.F. Zhang. "Research and Analysis of Operation and Maintenance Data Collection Technology in Power Communication System," Data Mining, 2018, 8(3):9. DOI: 10.12677/HJDM.201.83013

[12] DevOps-Model, Alipay, https://github.com/codefuse-ai/CodeFuse-DevOps-Model, 2024.

[13] H. Guo, J. Yang, J. Liu, L. Yang, L. Chai, J. Bai, J. Peng, X. Hu, C. Chen, D. Zhang, X. Shi, T. Zheng, L. Zheng, B. Zhang, K. Xu and Z. Li, "OWL: A Large Language Model for IT Operations," ArXiv abs/2309.09298, 2023.

[14] P. Lewis, E. Perez et al., "Retrieval-augmented generation for k nowledge-intensive nlp tasks," Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.

[15] J. White, Q. Fu, S. Hays, M. Sandborn, C. Olea, H. Gilbert, A. Elnashar, J. Spencer-Smith, and D. C. Schmidt, "A prompt pattern catalog to enhance prompt engineering with chatgpt," arXiv:2302.11382, 2023.

[16] Langchain-Chatchat, Chatchat-Space, https://github.com/chatchat-space/Langchain-Chatchat, 2024.

[17] QAnyting, Netease-youdao, https://github.com/netease-youdao/QAnything, 2024.

[18] GLM-4-9B, Team GLM, https://github.com/THUDM/GLM-4, 2024.

[19] QWenLM, Alibaba Cloud, https://github.com/QwenLM/Qwen, 2024.

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